

Trading Performance, Disposition Effect, Overconfidence, Representativeness Bias, and Experience of Emerging Market Investors

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ABSTRACT

Using brokerage account data from China, we study investment decision making in an emerging market. We find that Chinese investors make poor trading decisions: the stocks they purchase underperform those they sell. We also find that Chinese investors suffer from three behavioral biases: (i) they tend to sell stocks that have appreciated in price, but not those that have depreciated in price, consistent with a disposition effect, acknowledging gains but not losses; (ii) they seem overconfident; and (iii) they appear to believe that past returns are indicative of future returns (a representativeness bias). In comparisons to prior findings, Chinese investors seem more overconfident than U.S. investors (i.e., the Chinese hold fewer stocks, yet trade very often) and their disposition effect appears stronger. Finally, we categorize Chinese investors based on proxy measures of experience and find that “experienced” investors are not always less prone to behavioral biases than are “inexperienced” ones. Copyright © 2007 John Wiley & Sons, Ltd.

KEY WORDS disposition effect; investor behavior; overconfidence; representativeness bias

This study considers the extent to which Chinese investors make poor trading decisions and exhibit three particular behavioral biases: (a) the disposition effect, (b) overconfidence, and (c) the representativeness bias. These cognitive errors are forms of heuristic simplification, which stem from the brain’s tendency to make mental shortcuts rather than engaging in longer analytical processing. The tendency for people to suffer from these biases is well-established in the psychology and behavioral science literature (e.g., Grether, 1980; Joseph, Larrick, Steele, & Nisbett, 1996; Kahneman & Tversky, 1973; Lichtenstein, Fischhoff, & Phillips, 1982; Ritov, 1996; Tversky & Kahneman, 1974; Yates, 1990). Recently, the economics and finance literature

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also has found that U.S. investors suffer from these same behavioral biases (e.g., Barber & Odean, 2000, 2001; Odean, 1998a). In this study, we examine Chinese investors because Chinese culture is very different from U.S. culture, and culture can breed overconfidence at varying levels (e.g., Yates, Zhu, Ronis, Wang, Shinotsuka, & Toda, 1989). Some evidence exists that suggests that people raised in Asian cultures exhibit more behavioral biases than people from the United States (e.g., Yates, Lee, & Bush, 1997).

We also consider the degree to which trading performance and behavioral biases are related to five specific investor characteristics: investor's investing experience, age, trading frequency, personal wealth, and the location in which he or she resides. We specifically consider the possibility that experienced, middle-aged, active, and wealthy investors living in cosmopolitan cities are less prone to suffering from behavioral biases. To conduct our analyses, we analyze the trades of 46,969 individual investor brokerage accounts from a brokerage firm in China. Before we present our results, we discuss the behavioral biases examined in the paper, we highlight the cultural differences between China and Western countries, and then we describe the investor characteristics that we consider.

INVESTOR BEHAVIOR

Investors may be inclined toward various types of behavioral biases, which lead them to make cognitive errors. People may make predictable, nonoptimal choices when faced with difficult and uncertain decisions because of heuristic simplification. We test for the existence of trading errors and consider three behavioral biases, which are described as follows.

Overconfidence

One outcome of heuristic simplification (i.e., self-deception) occurs when people tend to think that they are better than they really are (Trivers, 1991). The psychology and behavioral science literature characterize people that behave as if they have more ability than they actually possess as being overconfident (e.g., Campbell, Goodie, & Foster, 2004; Lichtenstein et al., 1982; Yates, 1990). Investors who attribute past success to their skill and past failure to bad luck are likely to be overconfident. An investor who is overconfident will want to utilize his perceived superior ability to obtain large returns. Thus, overconfident investors trade often and underestimate the associated risks of active stock investing (e.g., Kyle & Wang, 1997; Odean, 1998b). When analyzing investor behavior using their stock brokerage data, trading frequency is commonly used as a proxy for overconfidence. Supporting these conjectures, Barber & Odean (2000, 2001) and Odean (1999) find that U.S. individual investors trade excessively, expose themselves to a high level of risk, and make poor ex post investing decisions (i.e., their investment portfolio returns are inferior to a benchmark market-wide portfolio return). Of course, trading frequency is a noisy measure of overconfidence. Investors with superior information and trading skills will utilize this ability by trading often to capture high returns. So people with actual high ability and people who wrongly believe they have high ability will both trade excessively. It is generally assumed that there are few truly high ability investors compared to the number of overconfident ones. Therefore, the trading frequency proxy is often believed to represent the behavior of overconfident investors on average.

Disposition effect

Consider the situation in which an investor wishes to sell a stock. The investor can sell a stock that has increased in price or one that has decreased in price. People avoid actions that create regret and seek actions

that cause pride. Nofsinger (2005) illustrates that selling the “winner” (the stock that has increased in price) validates a good decision to purchase that stock in the first place and stimulates pride. Selling the “loser” (the stock with a loss) causes the realization that the original decision to purchase it was poor, and thus stimulates regret. Shefrin & Statman (1985) state that the propensity to avoid regret and seek pride causes investors to be *predisposed* (thus the behavioral effect’s namesake) to selling winners too early and riding losers too long. This disposition effect is one implication of extending Kahneman & Tversky’s (1979) prospect theory to investment decision making. In prospect theory, the value function is concave in the area of gains and convex in the area of losses, implying risk aversion in the area of gains and risk seeking in the area of losses. An investment decision-making application of mental accounting is the process in which the mind keeps track of gains and losses on each stock held rather than at the portfolio level (Thaler, 1980). The disposition effect combines the prospect value function with the focus on individual stock gains and losses to predict that investors are relatively more willing to sell the winners than the losers in their portfolios. There may be times when selling the winner is the optimal wealth-maximizing decision. But this is generally not the case when stock price changes move in trends (called return momentum) and when capital gains are taxed. Interestingly, Shefrin and Statman also find that the disposition effect reverses in December when individual investors’ attention is more focused on minimizing capital gains taxes by engaging in tax-loss selling.

Consistent with the disposition effect, the economics and finance literature finds that investors often sell stocks that have performed well so that they can feel good about themselves. At the same time, investors tend not to sell their poorly performing stocks because they are not ready to acknowledge that they made a mistake and because they are afraid that the stocks may recover (i.e., they wish to avoid regret; Shefrin & Statman, 1985). Odean (1998a) finds that U.S. individual investors are more willing to sell stocks that have done well than those stocks that have done poorly. Grinblatt & Han (2005) derive an equilibrium model that explains how prospect theory and mental accounting (i.e., the disposition effect) creates a momentum return pattern. Frazzini (2006) empirically tests the model and concludes that when investors display the disposition effect, it induces a stock price underreaction to news announcements and a post-announcement price drift.

Representativeness bias

One mental shortcut, the representativeness bias, involves overreliance on stereotypes (Shefrin, 2005). Representativeness leads people to form probability judgments that systematically violate Bayes’s rule (see Grether, 1980; Kahneman & Tversky, 1973; Tversky & Kahneman, 1974). The representativeness bias has several implications to investment decision making. Investors may misattribute good characteristics of a company (e.g., quality products, capable managers, high expected growth) as characteristics of a good investment. This stereotype would induce a cognitive error as Lakonishok, Shleifer, & Vishny (1994) show that these “glamour” companies are often poor investments. Investors may also consider recent past returns to be representative of what they can expect in the future (DeBondt, 1993). Because of this extrapolation bias, investors might buy stocks that have recently increased in price (i.e., they have a myopic focus). Dhar & Kumar (2001) investigate the price trends of stocks bought by more than 62,000 households at a discount brokerage during a 5-year period. They find that investors prefer to buy stocks that had recently enjoyed some positive abnormal returns, consistent with the thinking that the past price trend is representative of the future price trend. In this study, we examine extrapolation bias.¹

¹Grinblatt & Keloharju (2000) find that foreign investors in Finland both buy past winners and sell past losers, thus showing trend-following in both purchases and sales. In their case, the findings might indicate rational behavior if this momentum trading strategy is shown to be profitable. For China, however, momentum trading is not known to be profitable, suggesting that an extrapolation bias is consistent with a cognitive error. We discuss this issue in more detail in the results section of our paper.

CULTURAL DIFFERENCES

The cognitive errors in investment decision making described in the previous discussion have been predominately identified in samples of U.S. individual investors, although individual investors from other Western cultures, including Finland (Grinblatt & Keloharju, 2000) and Israel (Shapira & Venezia, 2001) have also been examined. However, important psychological differences may exist between Chinese investors and investors in developed Western cultures. Cultural differences are frequently expressed in cognitive studies as an individualism–collectivism continuum (Hofstede, 1980). Asian cultures tend to be based on a more socially collective paradigm than Western cultures. For example, in Asian cultures, family or other group members will step in to help any group member who encounters a large catastrophic loss whereas in individualist Western cultures, a person making a risky decision is expected to bear personally the adverse consequences of his or her decisions. Collective-oriented societies allow for the social diversification of risky decisions in a similar manner to the purchase of an insurance policy. Therefore, because the impact of a catastrophic loss is different between Asian and Western cultures, the perception of this type of risk may also be different.

Additionally, because cognitive biases may be learned (Wolosin, Sherman, & Till, 1973), differences in cultural life experiences and education may cause differences in cognitive biases. For example, according to Yates et al. (1989), the Chinese education system encourages students to follow rather than criticize traditions and precedents whereas the American education system encourages students to challenge others' and their own opinions. Yates et al. suggest that this critical thinking style of Western cultures reduce the tendency to be overconfident. Although initial indications from the psychology literature suggest that Asian cultures have a higher degree of overconfidence than Western cultures, the literature is sparse. Weber & Hsee (2000) conclude: "The bottom line is that the topic of culture and decision making has not received a lot of attention from either decision researchers or cross-cultural psychologists" (p. 34). In studies of general knowledge conducted in China and Taiwan, Asians were found to be more overconfident than Americans (see Yates, Lee, & Shinotsuka, 1996; Yates et al., 1997). Studies of risk perception also find that people in Asian cultures (test groups were China, Japan, and Hong Kong) are less risk averse and more overconfident than people in Western cultures (test groups were United States, Germany, and Poland; see Kleinhesselink & Rosa, 1991; Keown, 1989; Weber & Hsee, 1998). In addition, Kim & Nofsinger (in press) report findings that Japanese investors were overconfident and less risk averse in the Japanese bubble market of the 1980s.

Less work has been done on the cultural implications of the disposition effect and the representativeness heuristic. Gilovich, Wang, Regan, & Nishina (2003) examine regret aversion in China and Japan and hypothesize that individuals in these more collectivist societies should show a different level of regret than compared to those in the individualist-oriented society of the United States. They compare their experiments' results with those of Gilovich & Medvec (1995), who conducted similar tests using American participants. In contrast to their hypothesis, Gilovich et al. find that regret is viewed nearly identically among the three cultures. Ji & Zhang (2006) examine the representativeness heuristic by contrasting the buy and sell behavior of Canadian and Chinese investors in three experiments. They find that Chinese investors are less prone to exhibit the extrapolation bias than Canadian investors. Indeed, Chinese participants were more likely to predict a price reversal than trend continuation.

INVESTOR CHARACTERISTICS

In our study, we also differentiate our results based on investor characteristics. We first contrast individual investors to institutional investors to see whether individuals are more prone than institutional investors to trading mistakes or biases. It is commonly assumed that institutional investors and individual investors behave differently. For example, the literature often considers individual investors less sophisticated and,

therefore, attributes irrational behavior and market anomalies to their trading. Ignorant, uninformed individual investors trading on sentiment is a common theme in the herding literature (see Nofsinger & Sias, 1999). Also, Shiller (1984) and DeLong, Shleifer, Summers, & Waldmann (1990) suggest that fad and fashion, rather than fundamentals, are likely to impact the investment decisions of individual investors.

Some compelling empirical support exists that sophisticated investors are more rational. Using data from Finland, Grinblatt & Keloharju (2000, 2001) find that sophisticated investors (which they believe to be the foreign institutional investors in their case) were more likely to follow momentum trading strategies (in Finland, momentum profits are significant, thus making momentum trading a rational trading strategy) and less likely to be inclined toward a home bias. However, the evidence about institutional investors behavior is mixed. To study the disposition effect, Shapira & Venezia (2001) examine brokerage accounts in Israel and find that, on average, individuals hold on to poorly performing stocks 8 days longer than do professional institutional investors. On the other hand, studies on professional U.S. futures traders (Coval & Shumway, 2005; Locke & Mann, 2005) and on U.S. mutual funds (e.g., Frazzini, 2006) find that institutional investors exhibit the disposition effect to a similar degree as individual investors.

If individual investors, compared with institutional investors, are more inclined toward trading mistakes or behavioral biases, on average, then an important follow-up question should be asked: Given that we would never expect all individuals to behave in the same way, what would explain why some individual investors behave one way while other individuals behave another way?² Psychologists find that different groups of people experience different levels of cognitive biases. For example, men seem to be more overconfident investors than women (Lundeberg, Fox, & Puncchar, 1994; Barber & Odean, 2001). Additionally, different experiences seem to lead to different behaviors (Wolosin et al., 1973; Gervais & Odean, 2001). Therefore, not all individuals might behave similarly when it comes to investing. We identify five investor characteristics that we believe (albeit perhaps naively) will be less prone to behavioral biases and trading mistakes. Specifically, we identify (a) experienced investors, (b) middle-aged investors, (c) active investors, (d) wealthier investors, and (e) investors from large cosmopolitan cities to see whether they are less inclined toward making cognitive errors in their investing decisions. We describe each investor characteristic in turn.

Experienced investors

We posit that investors who have held their brokerage account for a relatively longer period of time might be less inclined to making mistakes. These investors have accumulated investing experience, so they may have learned to become more rational. Investors who fail to learn and improve their skill over time may lose money and leave the market. Thus, experience may also represent a survivorship bias (however, there are some researchers that believe irrational investors can survive).³ List (2003) provides some experimental evidence in support of the theory that traders can learn to become rational. Korniotis & Kumar (2006) study the trade-off between the benefits of investing experience with the problems of cognitive aging. More experienced investors hold less risky portfolios, are better diversified, and trade less frequently. However, after controlling for experience, older investors have worse stock selection abilities and poorer diversification skills.

²We could also ask about explaining differences among institutional investor behaviors. Shefrin (2005) suggests that institutional investor behavior is heterogeneous. However, we only have data on 212 institutional investor accounts, and we also find limited variability in institutional investor characteristics (e.g., institutions do not have an age in the same way as individuals do, most institutional investors have large accounts, most are located in cosmopolitan cities, etc.). Thus, although we can contrast individual investors to institutional investors, we are limited in our ability to conduct cross-sectional tests on institutional investor behavior.

³See Kogan, Ross, Wang, & Westerfield (2006) and references therein.

Middle-aged investors

In China, younger people tend to be the more educated and willing to participate in capital market activities. However, older people have more life experience. Therefore, the most sophisticated investors are likely to be young enough to have a market-oriented education but old enough to have accumulated and learned from “life’s lessons”. Economic reforms in China began in 1978. A citizen who was 20 years old when the reforms began would be 40 years old at the beginning of our sample period. Therefore, we measure one’s sophistication as a function of age using the absolute value of 40 minus the investor’s age (i.e., $|40 - \text{investor's age}|$, where if the investor is 35 or 45 years old, then this variable is equal to 5 for either case). A small number indicates an investor with a good mix of a capital-markets education and accumulated wisdom.

Active investors

The more often an investor trades, the quicker he or she gains trading experience. As previously mentioned, experienced traders may be less inclined toward behavioral biases in their trading decisions. On the other hand, Barber & Odean (2000) assume that investors who trade more suffer from overconfidence. They find that investors who trade more achieve worse performance. Thus, active trading could be a sign of either an investor who has learned to be more rational or one who is overconfident.

Wealthier investors

Wealthier individuals may be more knowledgeable about finances than other individuals. However, it is also possible that wealthier individuals may be more overconfident. Vissing-Jørgensen (2003) studies survey data of U.S. investors from UBS/Gallup on stock holdings and expectations. She finds that although these investors suffered from several psychological biases (including the representativeness bias), having higher levels of wealth diminished these biases somewhat. We use the value of the equity in the brokerage account as a proxy for an investor’s wealth.

Investors from large cosmopolitan cities

Accounts are located in six different cities: Wuhan, Urumuchi, Futian, Shekou, Shanghai, and Pudong. Futian and Shekou are communities of Shenzhen, and Pudong is a community of Shanghai. Among the cities in our study, Shanghai and Shenzhen are the most cosmopolitan. Many of the major Chinese firms and many of the elite Chinese universities are located in Shanghai. As such, the overall technology and education levels are higher in Shanghai than in most other parts of China. Shenzhen, the other city defined as cosmopolitan, was the first Special Economic Zone in China. It was created as an experimental city where capitalism was allowed to flourish. Today, Shenzhen is considered one of the main economies in China. Located in the center of China, Wuhan is one of the important industrial cities in China. It is also a transportation hinge (nerve center) for both railway and shipping (e.g., the famous Yangtse River passes through it). Because Wuhan has good universities and some experience with business, its residents are considered moderately sophisticated. Urumuchi is in Xinjiang, which is a much more rural province in the far northwest part of China. The region is not known for the existence of large companies or good education institutions. Therefore, investors who have accounts in Shanghai and Shenzhen may be better investors than those from the more rural parts of China such as Urumuchi.

Our study in relation to other similar studies

Other economics and finance scholars have begun studying the characteristics and behavior of investors. Dhar & Zhu (2006) consider demographic and socioeconomic variables on the degree to which the disposition effect is demonstrated. They report that wealthier individuals and those working in professional occupations exhibit a lower disposition effect. Higher trading frequency is also associated with reduced disposition effect.

Feng & Seasholes (2005) study the impact of investor experience on the disposition effect in China. Our main distinction from these works is that they are more narrowly focused on one cognitive error (the disposition effect) whereas we employ a single dataset to study several cognitive biases simultaneously. In general, we believe this imposed uniformity is a major strength of our study. For example, our overall understanding of investor behavior comes from many separate studies of disparate, and often unrelated, investors. In addition, and unlike Dhar and Zhu and Feng and Seasholes, our study also compares individual investors with institutional investors. Therefore, we make important contributions to both the psychology and economics literature along several dimensions.

EMPIRICAL APPROACH OF THE STUDY

We conduct four sets of empirical tests on Chinese investors. First, we identify their investing performance by reporting the subsequent return of stocks they sold versus stocks they purchased. Second, we determine to what extent Chinese investors are disposed to holding poorly performing stocks (“losers”) while selling “winners” (i.e., the disposition effect). Third, we assess the extent to which they exhibit overconfident characteristics. Specifically, we examine their inclination toward portfolio diversification and trading activity. Fourth, we identify the past performance of stocks that investors are purchasing to see whether they are attracted to stocks that experience very recent run-ups (i.e., the extrapolation bias, a form of the representativeness bias). Given that we examine trading performance and three behavioral biases, we divide the discussion of our results into four separate subsections. An additional fifth subsection reports results which explores all biases simultaneously.

In conducting these tests on trading performance and behavioral biases, we take a three-step approach. First, for our sample of individual investors, we assess their trading performance and determine the existence of behavioral biases. Second, we compare our findings for individual investors with institutional investors. Third, we conduct cross-sectional regression tests on our sample of individual investors to see how individual investors with different characteristics behave.

DATA DESCRIPTION

The Chinese stock markets are composed of two securities markets, the Shanghai Stock Exchange (SHSE) and the Shenzhen Stock Exchange (SZSE), which were established in November 1990 and April 1991, respectively. These markets exhibited strong growth in the past decade and, when combined, are currently the ninth largest in the world. At present, about 1,250 companies are listed on the SHSE and the SZSE, with total market capitalization exceeding 4,000 billion Chinese yuan (about \$500 billion). Several important institutional differences also exist between China and the United States. In particular, investors in China may open only one brokerage account (Feng & Seasholes, 2003), which is opened using their National Identity Card. With only one brokerage account and lacking of 401(k) plans and other equity holdings, our study is likely viewing investors’ total investing behavior.

Our brokerage account data come from a brokerage firm in China.⁴ The complete dataset includes 74,960 investor accounts. Not all of these accounts are useful for our study. We first delete the 17,172 accounts that only own government bonds. We also delete 5,720 accounts that have a zero (or negative) balance, which leaves 52,280 individual investor accounts. We next delete 5,099 trading accounts that only participate in the primary market (i.e., the buying of initial public offerings), leaving a final sample of 46,969 individual investor accounts and 212 institutional investor accounts.

⁴As a condition for using its data, we promised anonymity to the brokerage firm.

Our sample period is from May 20, 1998 to September 30, 2002. For both individual and institutional investors, we have data on the number of years that the investor has held the account, the investor's trading activities (stocks bought and sold), the size of the investor's brokerage account, and the branch (city) in which the account is located. In addition, for individual investor accounts, we also have data on the individual's age. To calculate stock returns, we use the monthly returns files of the China Stock Market and Accounting Research (CSMAR) database, which is compiled and maintained at Hong Kong Polytechnic University. We omit returns that occurred during a stock's very first month of trading to eliminate any effects associated with new stock offerings made by companies.

Summary statistics of our brokerage account database are presented in Table 1. Panel A reports statistics for individual investors, and Panel B shows statistics for institutional investors. Panel A shows that our study includes 1,308,596 stock purchase orders and 1,091,848 stock sales for individual investor accounts. Panel A also shows that the average stock account in the individual investor sample has been open for 3 years, 4 months. Interestingly, the mean value of accounts open less than 3 years is greater than the mean value of accounts open longer than 3 years. Most of the individual investor accounts owners are between 29 and 49 years of age. The 10 percent of the individual account holders that exhibit the greatest amount of trading turnover also have an account value greater than twice the value of the other accounts. The currency exchange rate during this time period was roughly 8 Chinese yuan (CNY) to \$1. The mean value of the smallest third of accounts held by individuals, sorted by account value, is CNY7,023 (\$878). The mean account value in the largest third of the accounts is CNY307,737 (\$38,467). The greatest number of accounts held by individual investors is in Urumuchi, with 12,327 accounts; however, the two Shenzhen suburbs, Futian and Shekou, combine for a total of 17,262 individual investors. The greater Shanghai area has 12,608 individual accounts.

From Panel B of Table 1, in which we report summary statistics of the 212 institutional investors' accounts, we see that the mean account value of institutional investors is more than 76 times larger than the mean account value of individual investors (CNY8.68 million vs. CNY113,455). The institutional accounts have been open for 3 years, on average. The mean value of institutional investor accounts open more than 3 years is greater than the mean value of accounts open less than 3 years. The mean value of the largest one-third of accounts, sorted by account value, is much larger than the mean value of the smaller two-thirds of accounts. Perhaps not surprisingly, very few institutional investors are located in Urumuchi, a rural area.

TESTS AND FINDINGS

Investor characteristics and trading performance

Consider the investor who sells one stock and uses the proceeds to buy another stock. If the stock sold subsequently outperforms the stock bought, then the investor would have been better off holding the first stock. Making the trade was a poor decision, *ex post*. Following Odean's (1999) methods, we identify the average subsequent returns of stocks that investors purchase and sell. We compute the total return for each stock during the 4 months (84 trading days), 1 year (252 trading days), and 2 years (504 trading days) after the trade. Table 2 presents these returns.

Panel A of Table 2 shows that the subsequent return of the stocks purchased was 3.59% for the 4-month period following the trade. This finding compares with a return of 5.57% for the stocks sold. The mean difference for each individual (−1.40%) is significant at the 1% level. For 1 year and 2 years after the trades, the return difference between stocks purchased and stocks sold is −2.45% and −1.80%, respectively. Thus, we find that individual investors make poor trading decisions. Institutional investors in China, on the other hand, appear to make good trading decisions. The stocks they buy outperform the stocks they sell. For example, for 1 year and 2 years after the institutions' trades, the return difference between stocks purchased and stocks sold is about 5%. Therefore, our findings are consistent with the notion that individual investors, compared with institutional investors, are more prone to trading mistakes.

Table 1. Descriptive statistics on brokerage accounts and trades

Panel A. Individual investors						
	Number of accounts	Mean value of accounts	Mean yrs. account is open	Number of purchases	Number of sells	Mean transaction volume
All accounts	46,969	113,455	3.33	1,308,596	1,091,848	2,538
By account age:						
Open <3 years	16,789	135,132	1.65	298,422	230,506	2,899
Open ≥3 years	19,984	116,874	4.74	689,128	587,867	2,554
By age of investor:						
18–28 years old	4,682	121,265	2.31	89,553	75,314	2,284
29–49 years old	31,391	101,272	3.49	857,417	717,814	2,516
50–75 years old	2,437	73,765	3.38	76,322	62,743	1,965
By trading activity:						
Frequent trading	4,694	224,959	4.18	639,578	572,459	2,763
Infrequent trading	42,275	101,075	3.24	669,018	519,389	2,309
By account value:						
Small	15,656	7,023	3.07	200,783	177,167	616
Medium	15,657	25,612	3.36	369,631	307,735	1,051
Large	15,656	307,737	3.57	738,182	606,946	3,827
By city:						
Wuhan	4,772	66,365	2.50	146,197	123,604	2,051
Urumuchi	12,327	56,001	3.08	242,174	197,112	1,518
Futian (Shenzhen)	5,271	313,712	3.07	156,788	135,211	5,676
Shekou (Shenzhen)	11,991	101,455	4.59	394,261	337,092	2,604
Shanghai	8,175	108,903	3.28	257,334	210,265	1,927
Pudong (Shanghai)	4,433	126,656	2.52	111,842	88,564	2,042

Continues

Table 1. Continued

Panel B. Institutional investors					
	Number of accounts	Mean value of accounts	Mean yrs. account is open	Number of purchases	Number of sells
All accounts	212	8,678,648	2.96	88,981	59,458
By account age:					
Open <3 years	79	7,431,729	1.65	17,359	11,183
Open ≥3 years	118	9,788,379	3.85	71,622	48,275
By trading activity:					
Frequent trading	43	27,382,500	3.41	82,144	53,306
Infrequent trading	169	3,633,530	2.85	16,171	11,960
By account value:					
Small	65	121,450	3.03	1,929	1,518
Medium	65	1,235,792	3.02	11,796	10,072
Large	63	25,186,640	2.79	79,710	51,466
By city:					
Wuhan	12	719,035	2.82	1,842	1,625
Urumuchi	6	372,338	2.67	651	470
Futian (Shenzhen)	64	14,413,304	3.44	42,816	30,017
Shenkou (Shenzhen)	36	8,595,813	3.61	17,223	11,705
Shanghai	65	5,974,132	2.67	19,508	11,543
Pudong (Shanghai)	29	9,501,465	2.10	16,275	9,906

Note: This table reports descriptive statistics of Chinese individual investors' (Panel A) and institutional investors' (Panel B) brokerage accounts and their buy/sell trades during the period from May 20, 1998 to September 30, 2002. The data is also differentiated by how long the account has been open (open <3 years versus open ≥3 years), by the investor's age, by trading activity (frequent traders are those whose trading activity puts them in the top 10% of the sample), by account value in Chinese yuan (sample sorted into thirds), and by the city in which the account is located.

Table 2. Returns of stocks purchased versus stocks sold

Panel A: Univariate results			
Individual Investors			
	4-month returns	1-year returns	2-year returns
Purchases	0.0359	0.1124	0.1064
Sales	0.0557	0.1371	0.1243
Purchase–Sales	−0.0140*** (−182.58)	−0.0245*** (−90.49)	−0.0180*** (−41.87)
Institutional Investors			
Purchases	0.0193	0.0252	0.0468
Sales	−0.0178	−0.0362	0.0103
Purchase–Sales	0.0363*** (3.11)	0.0400*** (5.05)	0.0450*** (4.97)
Panel B: Regression results for individual investors			
	Dependent variable: (Purchase–Sales) for:		
	4-month returns	1-year returns	2-year returns
Intercept	−0.0060** (−2.50)	−0.0548*** (−17.83)	−0.2131*** (−36.42)
Account Age	0.0038*** (11.00)	0.0052*** (11.82)	0.0181*** (21.58)
40–Investor's Age	0.0002 (1.37)	0.0007*** (3.49)	0.0009** (2.27)
Frequent Trading Dummy	0.0031 (1.41)	0.0312*** (10.98)	0.0560*** (10.34)
Account Value	−0.0059*** (−6.34)	−0.0036*** (−3.10)	−0.0039* (−1.76)
Urumuchi (rural location)	−0.0036** (−2.09)	−0.0058*** (−2.56)	−0.0104** (−2.46)
Adjusted R ²	0.0081	0.0147	0.0314

Note: Panel A reports the average subsequent returns of stocks that investors purchased and sold from May 20, 1998 to September 30, 2002. The difference in subsequent returns between purchased stocks and sold stocks is also calculated, with a difference-of-mean *t*-test reported in parentheses. Panel B presents parameter coefficients of the following regression model: Purchase–Sales = $\alpha + \beta_1(\text{Account Age}) + \beta_2(|40 - \text{Investor's Age}|) + \beta_3(\text{Frequent Trading Dummy}) + \beta_4(\text{Account Value}) + \beta_5(\text{Urumuchi})$. The dependent variable is the difference between Purchase returns and Sales returns. Account Age is the number of years the account has been open, |40–Investor's Age| is the absolute value of 40 minus the investor's age, Frequent Trading Dummy is a dummy variable that indicates when the account is in the top 10% with regards to trading activity, Account Value is the equity value of the brokerage account, and Urumuchi is a dummy variable that indicates when the accounts are located in the rural city of Urumuchi. *t*-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1, 5 and 10 percent levels, respectively.

To examine the role that individuals' personal characteristics have on their trading performance, we regress personal characteristics onto the difference in purchase and sales returns. Specifically, the regression equation is

$$\begin{aligned}
 &\text{Purchase} - \text{Sales Return} \\
 &= \alpha + \beta_1(\text{Account Age}) + \beta_2(|40 - \text{Investor's Age}|) + \beta_3(\text{Frequent Trading Dummy}) \\
 &\quad + \beta_4(\text{Account Value}) + \beta_5(\text{Urumuchi}).
 \end{aligned} \tag{1}$$

The sample is the individual investor accounts. The dependent variable is the difference between the returns of stocks purchased and stocks sold using a subsequent 4-month time period, a 1-year time period, and

a 2-year time period. Account age is the number of years the account has been open, $|40 - \text{investor's age}|$ is the absolute value of 40 minus the investor's age,⁵ frequent trading dummy is a dummy variable that indicates when the account is in the top 10% with regard to trading activity, account value is the equity value of the brokerage account, and Urumuchi, the most rural city in our sample, is a dummy variable that indicates when the account is located there.

Panel B of Table 2 shows the regression estimates of equation (1). The reported adjusted R^2 for the regressions show that between 0.8% and 3.1% of the variance in the return difference is explained by the individual investor's personal characteristics. Although the adjusted R^2 s are low, they are comparable to other studies that conduct similar tests (e.g., Dhar & Zhu, 2006). The positive coefficient for the account age variable suggests that the longer a brokerage account has been open, the less stocks sold outperform stocks purchased. In this case, experience seems to lead to better investing performance. A positive coefficient for the investor age function is also found. However, this finding implies the opposite about investor sophistication: that is, middle-aged individuals are not, somewhat surprisingly, better investors.

The positive coefficient for the high trading frequency dummy variable means that high trading accounts perform better. The negative coefficient for the account value variable means that the largest accounts suffer poor trading performance the most. As previously mentioned, the residents of the city of Urumuchi are considered to have less experience with capitalism than the residents of Shenzhen and Shanghai. The negative coefficient on the dummy variable means that investors in that city have worse trading performance.

Overall, our results indicate that Chinese individual investors are making trading mistakes. More important, regression findings suggest that individual investors who are near age 40 and who have a higher account value are making poor trading decisions. In other words, middle-aged and wealthier investors are not better investors. However, Chinese individual investors who have held an account longer, trading in the top 10% of frequency, and not living in rural Urumchi seem to make fewer mistakes.

In comparison with other published studies, Odean (1999) studies U.S. individual investors and finds the difference between stocks purchased and sold to be -1.36% , -3.31% , and -3.32% , for the 4-month, 1-year, and 2-years periods after the trades, respectively. Thus, the difference between bought and sold stocks in China, compared with the United States, are similar in the short-term (4 months) and smaller in the long-term (1 and 2 years). Overall, however, it appears that Chinese individual investors are making poor trading decisions similar to U.S. individual investors.

Similar to our findings on trading activity, Odean (1999) also finds that trading frequency correlates with returns. Specifically, he finds that low-trading U.S. accounts have a purchase–sale return difference of -4.28% for the year following the trades and that high-trading accounts experience a purchase–sales difference of only -2.69% . The coefficient on the frequent trading variable in our regression suggests that the difference in China is 3.12% (vs. 1.59% in the United States). However, our finding suggests that investors who trade more earn higher returns, which is not consistent with Barber & Odean's (2000) finding that high volume traders underperform.

Investor characteristics and the disposition effect

When an investor buys a stock and the price subsequently declines, it is considered a “paper loss” until the stock is sold at the decreased price in which case it becomes a “realized loss.” Investors are reluctant to realize their losses (Grinblatt & Keloharju, 2001; Odean, 1998a). When stocks are sold at a higher price than which they were originally bought, we compute the proportion of gains realized to the total gains that *can* be

⁵Account age and investor age relative to 40 is possibly confounding in our regression model. However, we find that both investor age and age relative to 40 are not that highly correlated with the account age variable (correlation coefficient is 0.16 and 0.18, respectively). Furthermore, a variance-inflation factor test suggests that a confounding effect between the two variables is not serious in all of our regression models.

realized (i.e., realized gains plus unrealized paper gains). Specifically, for stocks sold at a capital gain, we compute a proportion of gains realized (PGR) measure where

$$\text{PGR} = \frac{\text{Realized Gains}}{\text{Realized Gains} + \text{Paper Gains}}. \quad (2)$$

Similarly, for stocks sold at a capital loss, a PLR measure is computed as the proportion of losses realized compared to the total loss that can be realized as follows:

$$\text{PLR} = \frac{\text{Realized Losses}}{\text{Realized Losses} + \text{Paper Losses}}. \quad (3)$$

A large PGR relative to PLR indicates that investors tend to sell their winners more than their losers.⁶

We compute the PGR and PLR for each sale in the sample. Table 3, Panel A reports the mean PGR, PLR, and the difference between PGR and PLR for all transactions. From Panel A, we see that the PGR for individual investors, at 0.519, is 0.209 larger than the PLR for individual investors, at 0.310. The difference is statistically significant at the 1% level. These results suggest that Chinese individual investors are reluctant to realize their losses.

Institutional investors in China, on the other hand, appear to be less affected than Chinese individual investors by the disposition effect. For institutional investors, the difference between PGR and PLR is only 0.0877, which is more than half the difference between the PGR and PLR difference for Chinese individual investors. Therefore, although institutional investors in China appear to exhibit the disposition effect, it is not as strong as the disposition effect exhibited by individual investors.

To disentangle the multivariate effects of individual investors' personal characteristics, we re-estimate regression (1), but this time the dependent variable is PGR, PLR, or the difference (PGR–PLR). Panel B of Table 3 reports the coefficient estimates. The first column of results, where PGR is the dependent variable, shows that individuals with older and larger accounts, and who trade often, are less inclined to recognize gains, whereas middle-aged individuals living in cosmopolitan cities are more inclined to recognize gains. The second column of results, where PLR is the dependent variable, shows that individuals with older and larger accounts are less inclined to realize losses, whereas middle-aged investors living in cosmopolitan cities are more inclined to realize losses. The third regression, where PGR–PLR difference is the dependent variable, shows the disposition effect (i.e., the larger the difference, the greater is the individuals' disposition to sell winners and hold losers). The regression results suggest investors who trade often and investors who have larger accounts suffer *less* from a disposition effect. However, middle-aged investors and investors from cosmopolitan cities seem to suffer *more* from a disposition effect.

Overall, we see that Chinese investors are inclined toward a disposition effect. That is, they are more likely to realize paper gains than paper losses. The third regression in Panel B of Table 3 shows that investor sophistication or experience either mitigates or exacerbates the individual investor's net disposition effect, depending on the measure of sophistication or experience. It is particularly noteworthy that middle-aged investors, who were previously found to make poor trading decisions, also suffer from a disposition effect.

As an additional test of the disposition effect, we examine the average length of time it takes investors to sell a stock with capital gains versus the time it takes to sell a stock with capital losses. If investors are more

⁶Shefrin & Statman (1985) predict that the disposition effect would reverse in December as investors become focused on taxes and tax-loss selling. However, because there are no capital gains tax in China, we do not expect to find a seasonality effect akin to what Shefrin and Statman find. In unreported tests, we find that the disposition effect is nearly equal when comparing the period of January through November to the month of December. In fact, we find that the disposition effect is slightly stronger in December for individual investors, primarily because they are slightly less likely to realize losses in December. For institutional investors, the disposition effect is the same across all months. Overall, these results are consistent with our expectation that the calendar effect for our disposition results would not be significant.

Table 3. The disposition effect: the propensity to realize gains and losses

Panel A: Univariate results			
	PGR	PLR	PGR–PLR (t-stat)
Individual investors	0.5190	0.3098	0.2092*** (82.60)
Institutional investors	0.6870	0.5990	0.0877*** (3.977)
Panel B: Regression results for individual investors			
	Dependent variable:		
	PGR	PLR	PGR–PLR
Intercept	0.7007*** (134.40)	0.4662*** (78.01)	0.2345*** (37.79)
Account Age	–0.0271*** (–15.42)	–0.0301*** (–14.95)	0.0030 (1.44)
40–Investor Age	–0.0016*** (–13.52)	–0.0010*** (–7.52)	–0.0005*** (–4.12)
Frequent Trading Dummy	–0.0747*** (–14.16)	0.0061 (1.01)	–0.0808*** (–12.88)
Account Value	–0.0864*** (–41.94)	–0.0771*** (–32.65)	–0.0092*** (–3.78)
Urumuchi (rural location)	0.0080* (1.96)	0.0316*** (6.74)	–0.0236*** (–4.84)
Adjusted R ²	0.1297	0.0726	0.0117

Note: Panel A reports the mean proportion of gains and losses that are realized for stock transactions that took place from May 20, 1998 to September 30, 2002. PGR is the ratio of realized gains to the sum of realized gains and paper gains. PLR is the ratio of realized losses to the sum of realized losses and paper losses. We also report the difference between PGR and PLR, along with a t-statistic indicating statistical significance. Panel B presents parameter coefficients of the following regression model: PGR (or PLR or $PGR-PLR$) = $\alpha + \beta_1(\text{Account Age}) + \beta_2(|40-\text{Investor's Age}|) + \beta_3(\text{Frequent Trading Dummy}) + \beta_4(\text{Account Value}) + \beta_5(\text{Urumuchi})$. Account Age is the number of years the account has been open, |40–Investor's Age| is the absolute value of 40 minus the investor's age, Frequent Trading Dummy is a dummy variable that indicates when the account is in the top 10% with regards to trading activity, Account Value is the equity value of the brokerage account, and Urumuchi is a dummy variable that indicates when the accounts are located in the rural city of Urumuchi. *t*-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1, 5 and 10 percent levels, respectively.

likely to sell a winner stock than a loser stock, then the realized holding period of stocks sold at a loss would be longer than that of stocks sold at a gain. We follow Shapira & Venezia's (2001) approach and compute the time duration (in days) of each round trip. A "simple" round trip is one purchase of a stock followed by the sale of all shares of that stock. Other round trip trades are more complicated. For example, an investor could buy 100 shares of a stock and then 200 shares on two different days and then sell all 300 shares on another day. Although we could report this more complicated trade as two different trades, we choose to report it as only one. The duration of the reported trade is the weighted average time from the first buy to the sale and from the time from the second buy to the sale. The weights used are the value (shares $\times$ price) of each buy.

The average duration of these round trip trades is reported in Table 4. The first set of results reports the full sample for all round trip deals. Individual investors take over 35 days to sell a losing position and only 26 days to sell a winning position. The difference of 9 days is statistically significant and is consistent with the disposition effect.

Table 4. Duration of buy/sell round trips in days

	Individual investors			Institutional investors		
	Losing Stock	Winning Stock	Difference	Losing Stock	Winning Stock	Difference
All Round Trip Trades						
Mean	35.60	26.19	9.28***	25.86	15.59	10.27***
Stddev	59.77	44.92		43.41	29.13	
# obs.	268,390	513,577		1,299	1,904	
Simple Trades						
Mean	30.21	22.79	7.42***	16.52	9.59	6.93***
Stddev	57.27	42.89		37.94	21.96	
# obs.	184,402	379,805		697	1,099	
Complex Trades						
Mean	48.17	36.32	11.85***	36.67	23.79	12.89***
Stddev	63.20	48.85		46.75	35.12	
# obs.	83,988	133,772		602	805	

Note: This table shows the average number of days it takes investors to sell a stock with capital losses (Losing Stocks) versus those with capital gains (Winning Stocks). A “simple” round trip trade is where there is one purchase of a stock followed by the sale of all shares of that stock. A “complex” round trip trade occurs when there was more than one purchase day. That is, the investor accumulates the stock position over multiple days. In this complex case we report the weighted average number of days from each buy to the sale. The weights used are the value (shares $\times$ price) of each buy. The difference in the number of days it takes to sell losers versus the number of days it takes to sell winners is also reported. *** denotes that the difference is statistically significant at the 1 percent level.

Table 4 also reports the duration of round trip trades for institutional investors. The durations for institutional trades are shorter than for individuals. For example, institutions average nearly 26 days to sell losers, compared with 35 days for individuals. Institutions average nearly 16 days to sell a winner, compared with 26 days for individuals. Notice that the difference between the loser and winner durations is similar for institutions and individuals.

Finally, we conduct a third test of the disposition effect. Recently, Feng & Seasholes (2005) point out that PGR and PLR measures might be problematic at the individual account level. First, these measures might be mechanically linked to various measures of experience (see Appendix D of their paper for a numerical illustration); and, second, these measures may not be smooth, continuous variables at the individual level (e.g., some investors may only sell stocks for gains or only sell stocks for losses). Therefore, they suggest a survival analysis to test how long an investor holds a stock before selling it. A survival analysis has several advantages: (a) it can offer a baseline statistical model for how long stocks are typically held, (b) it uses data on days investors do not buy/sell, and (c) it is easy to interpret. For example, Feng and Seasholes show in their data that half the stocks purchased are sold within 10 days, regardless of whether they have increased or decreased in price. Therefore, computations of the probability that a stock is sold because it is a winner or loser needs to be compared with this general baseline probability of a sale.

The survival analysis method is as follows. For every investor and for every day of the sample period, a dependent variable takes the value of 1 for every day that a stock is sold, and a value of zero for every day the stock is held. Two independent variables are created: a trading loss indicator (TLI) variable, which takes the value of 1 when the stock is trading at a loss, and zero otherwise; and a trading gain indicator (TGI) variable, which takes the value of 1 when the stock is trading at a gain, and zero otherwise. The parameter coefficients that we report are hazard ratios for the average individual's decision to sell/hold stocks for a loss/gain.⁷ A TLI parameter coefficient of less than 1 suggests that investors are less likely to sell for a loss than what a baseline

⁷We use a Weibull distribution with parameter p to parameterize the hazard function. In all of our regression models, the p -parameter is less than 1 and is statistically significant, indicating a decreased hazard rate over time. See Feng & Seasholes (2005) for more technical details of the hazard ratio methodology.

hazard function would suggest. Note that the baseline hazard function represents the general probability of a sale given the number of days since the purchase, irrespective of any capital gains or losses. A TGI parameter coefficient of greater than 1 suggests that investors are more likely to sell for a gain than what a baseline hazard function would suggest. Both findings are consistent with the disposition effect. Table 5, Panel A, shows the regression results.

Table 5. Hazard ratio tests of the disposition effect

Panel A: Simple indicator regression models						
	Individual investors			Institutional investors		
TLI	0.653***			0.685***		
TGI		1.861***				0.998
Panel B: Expanded regression with TLI interactions (for individual investors)						
Disposition Effect Variables						
TLI	0.547***	0.656***	0.770***	0.661***	0.655***	0.803***
Account Age $\times$ TLI	1.408***					1.028***
40–Investor Age $\times$ TLI		1.098***				1.002
Frequent Trading Dummy $\times$ TLI			0.798***			0.747***
Account Value $\times$ TLI				1.003		1.000
Urumuchi (rural location) $\times$ TLI					1.104***	1.040
Control Variables						
Account Age	1.036					1.111***
40–Investor Age		1.103***				1.218***
Frequent Trading Dummy			0.853***			0.993
Account Value				1.000		1.000
Urumuchi (rural location)					1.210***	1.187***
Panel C: Expanded regression with TGI interactions (for individual investors)						
Disposition Effect Variables						
TGI	1.784***	1.903***	1.875***	1.803***	1.803***	1.802***
Account Age $\times$ TGI	0.961					0.918***
40–Investor Age $\times$ TGI		0.918				0.980***
Frequent Trading Dummy $\times$ TGI			1.175*			1.323***
Account Value $\times$ TGI				0.981		1.000
Urumuchi (rural location) $\times$ TGI					0.814***	0.872***
Control Variables						
Account Age	1.321***					1.202***
40–Investor Age		1.169				1.085***
Frequent Trading Dummy			0.925			0.956***
Account Value				1.000		1.000
Urumuchi (rural location)					1.170**	1.204***

Note: For every investor and for every day of the sample period, a dependent variable takes the value of 1 for every day that a stock is sold, and a value of zero for every day the stock is held. Two independent variables are created: a trading loss indicator (TLI) variable, which takes the value of 1 when the stock is trading at a loss, and zero otherwise; and a trading gain indicator (TGI) variable, which takes the value of 1 when the stock is trading at a gain, and zero otherwise. All numbers reported in this table are regression parameter coefficients (hazard ratios). A TLI parameter coefficient of less than 1 suggests that investors are less likely to sell for a loss than what a baseline hazard function would suggest. A TGI parameter coefficient of greater than 1 suggests that investors are more likely to sell for a gain than what a baseline hazard function would suggest. ***, **, and * denote statistical significance at the 1, 5 and 10 percent levels, respectively.

For individual investors, the TLI parameter coefficient is 0.653, which is less than 1. Therefore, individuals are less likely to sell for a loss than what a baseline hazard function would suggest. That is, individuals tend to hold onto losers. We also see that the TGI parameter is 1.861, which is greater than 1. Thus, individuals are more likely to sell for a gain than what a baseline hazard function would suggest. That is, individuals are quick to realize gains. Overall, the evidence is consistent with our prior findings: Individuals seem to suffer from a disposition effect. Institutional investors also tend to hold onto losers, but they do not seem to suffer from a tendency to realize gains quickly. They appear more willing to let their winners run.

In Table 5, Panel B, we report parameter coefficients of expanded regression models in which we interact each of our individual investor characteristic variables with the TLI variable and include the interactions as independent variables. We also include the main effect variables (i.e., control variables) without the TLI interaction in the regression models. Our models test one investor characteristic variable at a time, whereas the last model (last column of results) tests all investor characteristic variables simultaneously. Panel B, shows that individuals with older accounts are more inclined to realize losses. However, middle-aged individuals living in cosmopolitan cities who trade frequently are less inclined to realize losses (i.e., they suffer more from a disposition effect than others). Overall, these TLI interaction results are consistent with our earlier PLR regression findings.

Table 5, Panel C, where we interact each of our individual investor characteristics with the TGI variable, shows that individuals with older accounts are less likely to sell to realize gains. However, we see that middle-aged individuals living in cosmopolitan who trade frequently are more likely to realize gains. These results are similar to our earlier PGR regression findings.

Overall, all three of our disposition tests show that Chinese investors suffer from a disposition effect, although individual investors suffer more than institutional investors. The disposition effect derives from prospect theory. But some people seem to behave more along the axioms of prospect theory (i.e., more willing to take risks to avoid losses than to achieve gains) than others. Frederick (2005) finds that people with a more intuitive thinking process have strong prospect theory behavior while analytical thinkers do not. Therefore, analytical thinkers may exhibit less disposition effect behaviors than intuitive behaviors. Our sample does not include a measure of intuitive/analytical thinking process. However, it is likely that institutional investors will use more analytical thinking processes on average than individual investors.

Our overall tests also reveal that sophisticated individual investors are not always better investors. In particular, middle-aged investors living in cosmopolitan cities seem the most effected by the disposition effect.

In comparison to other published studies, Wermers (2003) also finds that U.S. institutional investors (i.e., mutual funds) tend to hold on to losers. However, Wermers reports that it is mutual funds with poor performance that have a greater tendency to hold loser stocks. Frazzini (2006) shows that PGR–PLR difference for U.S. mutual funds is a significant 0.031. Although significant, it is lower than the difference Odean (1998a) reports for individual investors (0.050). Odean's finding is also consistent with Shapira & Venezia (2001), who also report that individual investors are more inclined to the disposition effect than institutional investors.

Odean's (1998a) specific mean estimates of PGR and PLR for U.S. individual investors are 0.148 and 0.098, respectively. Dhar & Zhu (2006) suggest that the disposition effect should only occur in the United States from January through November due to tax-loss selling in December. Excluding December, they report a PGR and PLR of 0.38 and 0.17, respectively. Note that the disposition test ratios are much higher for Chinese investors (refer to our Table 3, Panel A, results) than they are for U.S. investor accounts. Using Frederick's (2005) line of reasoning that intuitive thinkers suffer more from a disposition effect as compared to analytical thinkers, one could mildly speculate that Chinese individual investors (as compared to U.S. individual investors) may be relying more on intuition than on analytics when making investing decisions.

As mentioned previously, Gilovich et al. (2003) contrast regret aversion in China, Japan, and the United States and hypothesize that China and Japan, as more collectivist societies, should show a different level of regret aversion than Americans. Surprisingly, they find that regret aversion tendencies are nearly identically in all three countries. Our findings could be viewed similarly. For U.S. individuals, mean PGR is about 50% larger than PLR (0.148 vs. 0.098). For Chinese individuals, mean PGR is about 60% larger than PLR (0.519 vs. 0.310). However, we should be careful in reading too much into this comparison between China and the United States. The Chinese stock markets have only been open for a short period, which means that the proportion of unrealized gains and losses (i.e., the ratio denominator) is likely to be much smaller for Chinese accounts than for U.S. accounts. In other words, Chinese investors have had less time to accumulate substantial gains and losses. Nevertheless, note that the same disposition effect is illustrated in both Chinese and U.S. investors.

In Shapira & Venezia's (2001) round trip analysis, they report that Israeli individual investors sell losers and winners after 63 and 20 days, respectively. Compared with our findings, Chinese individual investors, who sell losers and winners after 35 and 26 days, respectively, appear to suffer the disposition effect to a lesser degree than Israeli investors. These conclusions hold true for the subsamples of simple deals and complicated deals.

Feng & Seasholes (2005) also study the disposition effect for Chinese individual investors using survival analysis and find that experience actually reduces the disposition effect. However, they use different measures of experience than this study. They use (a) the number of ways investors place orders (i.e., the more ways, the more experienced is the trader), (b) number of stocks held, (c) gender, and (d) age (similar to our study, they contend that middle-aged investors rather than older investors are more experienced). Among their variables, the only common "experience" variable between their disposition tests and this study is the age variable, but they create four dummy variables for age ranges whereas we treat age as a continuous variable. They report that investors in the age group from 35 to 45 years are both least likely to sell a loser and most likely to realize a gain.

Investor characteristics and overconfidence

Overconfident investors feel certain about their knowledge and their investment skills. This overconfidence causes them to (a) trade too much and (b) hold underdiversified portfolios (Odean, 1998b). To measure trade frequency, we follow Barber & Odean (2001) and compute the mean monthly portfolio turnover for each brokerage account. To examine the potential for underdiversified portfolios, we report the mean number of stocks in each account.

We also measure each account's mean market-adjusted return to get a sense of how the investors' portfolios are performing. To measure market-adjusted returns, we use the beginning-of-month position data for each account and obtain their monthly stock returns, net of the equally weighted market's return, from the CSMAR database (using a value-weighted market return yields similar findings). We assume that all transactions take place on the last day of the month. The monthly market-adjusted return of an account's portfolios, R_t^j , is calculated as follows:

$$R_t^j = \sum_{i=1}^{H_t^j} p_{it}^j R_{it}, \quad (4)$$

where p_{it}^j is the beginning-of-month market value for the holding of stock i in account j in month t divided by the beginning-of-month market value of all stocks held in account j , R_{it} is the monthly market-adjusted return for stock i at month t , and H_t^j is the number of stocks held in account j in month t .

Table 6. Overconfidence tests

Panel A: Univariate results			
	Mean # of stocks owned	Mean monthly turnover	Mean monthly abnormal returns
Individual investors	2.60	0.2731	0.0042
Institutional investors	15.46	0.4306	0.0053
Panel B: Regression results for individual investors			
	Dependent variable:		
	Mean # of stocks owned	Mean monthly turnover	Mean monthly abnormal returns
Intercept	1.4847*** (56.18)	0.3944*** (107.06)	−0.0062*** (−15.05)
Account Age	−0.0023 (−0.58)	0.0321*** (57.51)	0.0026*** (41.60)
40−Investor's Age	0.0007 (0.39)	−0.0004 (−1.29)	0.0000 (0.52)
Frequent Trading Dummy	1.9291*** (61.53)		0.0055*** (11.16)
Account Value	0.8714*** (78.44)	0.0038*** (3.69)	−0.0006*** (−3.81)
Urumuchi (rural location)	0.0026 (0.12)	0.0122*** (4.04)	−0.0001 (−0.54)
Adjusted R2	0.3016	0.1069	0.0636

Note: Panel A reports the mean number of stocks owned in each Chinese brokerage account, the mean monthly turnover of the accounts, and the mean monthly returns for the accounts for the period from May 20, 1998 to September 30, 2002. Panel B presents parameter coefficients of the following regression model: Dependent Variable = $\alpha + \beta_1(\text{Account Age}) + \beta_2(|40 - \text{Investor's Age}|) + \beta_3(\text{Frequent Trading Dummy}) + \beta_4(\text{Account Value}) + \beta_5(\text{Urumuchi})$. The Dependent Variable is either the mean number of stocks owned in each account, the mean monthly turnover, or mean monthly returns for the accounts. Account Age is the number of years the account has been open, |40−Investor's Age| is the absolute value of 40 minus the investor's age, Frequent Trading Dummy is a dummy variable that indicates when the account is in the top 10% with regards to trading activity, Account Value is the equity value of the brokerage account, and Urumuchi is a dummy variable that indicates when the accounts are located in the rural city of Urumuchi. *t*-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1, 5 and 10 percent levels, respectively.

Table 6, Panel A, reports the mean number of stocks held in each account, the mean monthly turnover of each account, and the total mean monthly return of stocks held in each individual investor's account. Several observations are noteworthy. First, we see that in general, Chinese individual investors own very few stocks. On average, they hold the stocks of only 2.60 different companies. Consider that Chinese individual investors can have only one brokerage account and do not have access to other ownership forms like pension plans. To compensate, Chinese individuals should maintain diversification within their brokerage accounts. However, this does not seem to be the case.⁸ Polkovnichenko (2005) suggests that people underdiversify purposely: They know the risks but attempt to “get ahead” by hoping to capture the unlikely extreme gains only possible in an underdiversified portfolio.

⁸The transactions costs of equity trading in China is about 1% of the total transaction and consists of the following components: (a) brokerage commission fees, which are equal to 0.3% of the money amount of the transaction or CNY5, whichever is larger, (b) stamp duty, which is equal to 0.3% of the money amount of the transaction or CNY5, whichever is larger, and (c) transfer of ownership fees, which is equal to 0.1% of the money amount of transaction. If Chinese individuals view a 1% percent transactions cost as large, then this fee might represent another reason why Chinese individual investors are undiversified. However, we do not believe that the transactions cost is what prevents Chinese individual investors from holding more stocks in their portfolios. In the following discussion, we show that Chinese individual investors are frequent traders.

Chinese individual investors also appear to be trading quite frequently. The mean monthly turnover is 27.3%, or 327% annually, which, when an investor holds only two stocks, equates to trading (sell and then buy) a stock every other month. The high trading volume may be due to the lack of other investment vehicles in China.⁹ With regard to portfolio returns, individuals realize less than 0.5% in abnormal monthly returns. We discuss these returns in more detail in our subsequent regression analysis.

Chinese institutional investors own more stocks than Chinese individual investors (mean 15.5 stocks vs. 2.6 stocks) and institutional investors trade more often than individuals, but their trading yields better returns (0.53% abnormal returns vs. 0.42% returns). These results indicate that institutional investors are more diversified and trade more often than individuals. These results could easily be viewed as unsurprising, as institutional investors are “professional traders”. Consistent with this view, their trading yields a positive 0.53% abnormal return.

To examine the role for which individuals’ personal characteristics have on their tendencies to be overconfident, we again estimate regression (1). This time we use our overconfidence metrics (i.e., number of stocks held, turnover, and portfolio returns) as dependent variables. Panel B of Table 6, where the coefficient estimates are presented, shows that investors with older accounts have higher turnover but also a higher return. The coefficient estimates for investor age show that age is invariant to overconfidence tendencies. Investors who trade very frequently (top 10% of turnover) own 1.9 more stocks, on average, and earn 0.5% more return per month. Larger accounts also hold more stocks but trade more and earn a lower return. Urumuchi accounts also trade more.

Overall, the findings require careful interpretation. Older accounts have more trading (an overconfidence characteristic) but also have better returns. Those who trade frequently are actually more diversified, but they also earn higher returns despite their frequent trading. Larger accounts show more diversification, but they experience higher turnover and they also earn less. The evidence suggests that when experience makes investors overconfident, the overconfidence seems to be justified *ex post*, as these overconfident investors seem to earn higher returns. Only the investors with large accounts earn less.

In comparison to other published studies, Dhar & Zhu (2006) report that U.S. individual investors hold an average of four stocks; thus, the average Chinese individual investor is more underdiversified than the average U.S. individual investor. Zhu (2002) reports the monthly turnover of U.S. individual investors is 7.59% (or 91% annually). The evidence that Chinese investors are trading at a rate almost four times higher than U.S. investors seems extraordinary. These comparisons between China and the United States are also consistent with the observations of Allen, Qian, & Qian (2005). They report that the total turnover (expressed as a percentage of total market capitalization) for 1999 was 500% in China. This turnover is more than five times higher than the turnover of 87.7% on the NYSE during the same year. Overall, because Chinese individuals hold fewer stocks and trade more often than U.S. individual investors, our findings suggest that Chinese individual investors are more overconfident than U.S. individual investors, which is consistent with the general contention that people of Asian cultures are more overconfident than people of Western cultures. Finally, although we find that Chinese investors who trade frequently are more diversified and earn higher returns, these findings contrast with Barber & Odean’s (2000) findings that high volume U.S. traders underperform.

Investor characteristics and the recent past performance of stocks purchased

Investors might consider recent past returns to be representative of what they can expect in the future. This heuristic simplification is called the *extrapolation bias*, which is a form of the representativeness bias.¹⁰

⁹Grinblatt & Keloharju (2005) provide an alternative interpretation of our trading activity as they suggest that sensation-seeking by individual investors also induces more trading.

¹⁰In some cases, people believe a trend will reverse. This is known as the “gambler’s fallacy”, also a representativeness bias (Shefrin, 2005).

Table 7 reports the past return for the stocks that Chinese investors purchased. Specifically, the prior 4-month return, prior 4-month abnormal return, prior 1-year return, and prior 1-year abnormal return are shown. Abnormal returns are again calculated as actual returns minus the market return. Positive past returns indicate that investors buy stocks that recently increased, and positive abnormal returns indicate that investors buy stocks that did better than the overall stock market.

Univariate analysis is shown in Panel A of Table 7. Overall, individual investors purchased past winners. The return 4 months before the purchases is a surprisingly high 17% (a 32% abnormal return). The past 1-year return was only 2.9% (a 0.4% abnormal return). It appears that investors focus on the most recent performance of the stocks they purchase; that is, investors are myopic. Panel A also shows that institutional investors are less myopic: The past 4-month returns of the stocks they purchased is less than 1%. In fact, when looking at the past 1-year past returns of stocks that institutions purchased, institutional investors appear less inclined to use past performance to make purchasing decisions. That is, the extrapolation bias of assuming that stocks' past performance are indicative of stocks' future performance seems only applicable to individual investors and not to institutional investors.

Table 7. Past performance of stocks purchased

Panel A: Univariate results				
	Past 4-month returns	Past 4-month abnormal returns	Past 1-year returns	Past 1-year abnormal returns
Individual investors	0.1722	0.3250	0.0286	0.0043
Institutional investors	0.0087	0.0022	0.0283	−0.0042
Panel B: Regression results for individual investors				
	Dependent variable:			
	Past 4-month abnormal returns		Past 1-year abnormal returns	
Intercept	−0.0149*** (−8.59)		−0.0993*** (−37.48)	
Account Age	−0.0023*** (−8.79)		−0.0034*** (−8.50)	
40−Investor Age	0.0007*** (5.64)		0.0010*** (5.52)	
Frequent Trading Dummy	0.0036 (1.84)		0.0128*** (4.28)	
Account Value	0.0087*** (12.14)		0.0192*** (17.51)	
Urumuchi (rural location)	−0.0018 (−1.34)		0.0089*** (4.29)	
Adjusted R ²	0.0094		0.0155	

Note: Panel A reports past mean raw returns and past mean abnormal returns of stocks that were purchased during the period from May 20, 1998 to September 30, 2002. Abnormal returns are raw returns net of market returns. Panel B presents parameter coefficients of the following regression model: Past Performance = $\alpha + \beta_1(\text{Account Age}) + \beta_2(|40 - \text{Investor's Age}|) + \beta_3(\text{Frequent Trading Dummy}) + \beta_4(\text{Account Value}) + \beta_5(\text{Urumuchi})$. The dependent variable is either the past 4-month abnormal return or the past 1-year abnormal return. Account Age is the number of years the account has been open, |40−Investor's Age| is the absolute value of 40 minus the investor's age, Frequent Trading Dummy is a dummy variable that indicates when the account is in the top 10% with regards to trading activity, Account Value is the equity value of the brokerage account, and Urumuchi is a dummy variable that indicates when the accounts are located in the rural city of Urumuchi. *t*-statistics are reported in parentheses. ***, **, and * denote statistical significance at the 1, 5 and 10 percent levels, respectively.

To examine the role that individuals' personal characteristics have on their tendencies to succumb to the extrapolation bias, we again estimate regression (1), but this time we use prior abnormal performance (4 months and 1 year) as dependent variables. The coefficient estimates are presented in Panel B of Table 7, which shows that investors with accounts open longer are less myopic (i.e., they do not buy recent winners), as are middle-aged investors. However, people who trade more and have larger accounts have a greater tendency to buy recent past winners. These results suggest that active and wealthy investors are more likely to buy stocks with recent run-ups (i.e., they are more myopic). Individuals from Urumuchi seem to focus on the past 1-year returns more than on the past 4-month returns, indicating that they are a little myopic, but not very myopic. If investors from Urumuchi are less sophisticated than investors from more cosmopolitan cities, then we would have expected Urumuchi investors to be very myopic. Again, the mixed regression results indicate that investors assumed less likely to suffer from behavioral biases are still inclined, if not more inclined, toward behavioral biases.

In comparison to other published studies, Jegadeesh & Titman's (1993, 2001) studies are particularly relevant. They show that buying stocks with high returns from the past 3 to 12 months (past winners) and selling stocks with low returns during the same past period (past losers) is a trading strategy that yields positive abnormal returns for the following 3 to 12 month periods. This finding is widely known as the *momentum effect*. Rouwenhorst (1998) finds that the momentum effect also exists in European countries. Therefore, the buying of past winners by U.S. investors (Bange, 2000; Odean, 1998a) might be viewed as rational if momentum profits exits.

However, for Asian countries, the findings on momentum effects are mixed. Chui, Titman, & Wei (2000) study eight Asian countries (not including China) and find that the existence of momentum profits vary with the particular country, the country's legal system, holding period horizons, firm characteristics such as market capitalization and book-to-market, foreign ownership, and the time periods studied. Kang, Liu, & Ni (2002), who conduct a thorough study on momentum trading strategies for the Chinese stock markets find that momentum profits are very weak—most of the time the returns are not statistically different from zero (see Kang et al., Tables 1 and 2). Therefore, buying stocks based past returns in China cannot necessarily be described as a rational trading strategy, *ex post*. Because Chinese individual investors tend to buy past winners yet underperform in general, the behavior can best be described as an irrational extrapolation bias.

Are investors with one bias prone to other biases?

So far, we have examined several investor characteristics and several psychological biases. This section investigates the extent to which more than one cognitive bias might impact investors. We are not aware of any work that suggests that investors suffering from overconfidence should or should not be more likely to suffer from another bias, such as the representativeness bias or the disposition effect. Thus, this investigation is exploratory in nature.

We begin this multi-bias analysis by identifying whether each investor account shows evidence of one or more of the four following cognitive errors: trading errors, disposition effect, overconfidence, and representativeness. Dummy variables for each of the four biases are assigned the value of 1 if the investor suffers from the bias, and a value of zero otherwise. The first variable (D1, trading errors) is created using the stock purchase minus stock sales return analysis of equation (1) and Table 2. D1 equals 1 if the average of the purchase minus sales 1-year return is negative for the investor account, and zero otherwise. The disposition effect dummy (D2) equals 1 when the average PGR–PLR difference (of Equations 2 and 3) is positive and the average losing stock duration is greater than the average winning stock duration (see Table 4). The overconfidence dummy variable (D3) is equal to 1 when both the average number of stocks held during the period is 3 or less and the mean monthly trading turnover is greater than 0.27. Note that these breakpoints align with the sample means shown in Table 6. Finally, D4, the extrapolation dummy, equals 1 when the average past abnormal 1-year return of stocks purchased is positive.

Table 8. Multi-bias analysis

Panel A: Means of dummy variables that equal 1 if investor suffers from behavioral bias				
	D1	D2	D3	D4
	(Trading errors)	(Disposition)	(Overconfidence)	(Extrapolation)
Individual investors	0.17	0.35	0.23	0.65
Institutional investors	0.28	0.43	0.26	0.70

Panel B: Correlations				
Individual investors				
D1	D1	D2	D3	D4
D1	1			
D2	−0.139**	1		
D3	0.030**	−0.155**	1	
D4	0.037**	0.085**	−0.151**	1
Institutional investors	D1	D2	D3	D4
D1	1			
D2	0.014	1		
D3	−0.028	−0.167	1	
D4	−0.008	0.026	−0.061	1

Panel C: Frequencies (% of sample)					
	0 Bias	1 Bias	2 Biases	3 Biases	4 Biases
Individual investors	13.37	43.81	37.72	5.00	0.09
Institutional investors	8.79	32.97	47.25	9.89	1.10

Note: Dummy variables for each of the four biases are assigned a value of 1 if the investor suffers from the bias, and a value of zero otherwise. D1 (trading errors) equals 1 if the purchased stocks' average subsequent 1-year return minus the sold stocks' average subsequent 1-year return is negative for the investor account (refer to Table 2). D2 (disposition effect) equals 1 when the average PGR–PLR difference is positive (refer to Table 3) and the average losing stock duration is greater than the average winning stock duration (refer to Table 4). D3 (overconfidence) is equal to 1 when both the average number of stocks held during the period is 3 or less and the mean monthly trading turnover is greater than 0.27 (refer to Table 6). D4 (extrapolation) equals 1 when the average past abnormal 1-year return of stocks purchased is positive (refer to Table 7). Panel A shows the mean value of the four cognitive error dummy variables. Panel B shows the correlations among the dummy variables. Panel C shows the frequency distribution of the number of cognitive biases exhibited in the brokerage accounts. ***, **, and * denote statistical significance at the 1, 5 and 10 percent levels, respectively.

Panel A of Table 8 shows the mean value of the four cognitive error dummy variables. The mean can be interpreted as the fraction of investors suffering from that particular bias. As defined, 17% of the individual investors make poor trading choices on average. In comparison, 28% of the institutional investors make poor trading choices. Of individual investors, 35% and 23% exhibit the disposition effect and overconfidence, respectively, and 65% exhibit the extrapolation bias. Seventy percent of institutional investors exhibit the extrapolation bias.

The tendency for multiple biases to occur in the same investor account is illustrated by correlations in Panel B of Table 8. For the individual investor sample, all correlation estimates are statistically significant. The results indicate that investors suffering from poor trading choices are less likely to suffer from the disposition effect (correlation = −0.14). Investors suffering from overconfidence are less likely to suffer from the disposition effect (correlation = −0.16) and extrapolation bias (correlation = −0.15). On the other hand, investors suffering from poor trading choices are more likely to also be impacted by overconfidence (correlation = 0.03) and extrapolation bias (correlation = 0.04), although these correlations are small.

Finally, investors that exhibit the disposition effect have a positive correlation (0.09) with the extrapolation bias. The correlation estimates for institutional investors are not statistically significant.¹¹

The last of our multi-bias analysis, which is reported in Panel C of Table 8, shows the frequency distribution of the number of cognitive biases exhibited in the brokerage accounts. The results show that 86.6% of individuals exhibit as least one bias. Nearly 43% of the accounts show evidence of more than one bias, but only 5% of the accounts exhibit three or four biases. The institutional accounts also appear to show a propensity toward having multiple biases, as more than 58% of institutional investor accounts show two or more biases; 11% of the institutional accounts show evidence of three or four biases.

Although no studies explicitly examine whether various investor biases are correlated, a few studies do provide some insights. In a sample of U.S. brokerage accounts, Goetzmann & Kumar (2005) examine whether people with underdiversified portfolios also suffer from trend following (i.e., the extrapolation bias, a form of representativeness bias) and find a positive relation. Benartzi (2001) illustrates that multiple biases can exacerbate an investor's behavior. He shows that employees invest too much of their 401(k) retirement money into their company's stock—an outcome of the familiarity bias. This problem is magnified when the company's stock has substantially increased in the past 5 years. Investors extrapolate this past return as an expected return and tend to own an even higher allocation of the company stock. However, although Benartzi shows that investors can suffer from multiple biases, he does not specifically investigate the presence of different cognitive errors.

CONCLUSIONS

Recent research finds that individual investors are inclined toward behavioral biases and that they make trading mistakes. However, are emerging market investors more inclined or less inclined toward behavioral biases and trading mistakes, as compared to developed market investors? To address this question, we study Chinese investors. In empirical tests, we find that Chinese investors make trading mistakes (i.e., the stocks they sell outperform the stocks that they buy), they are reluctant to realize their losses (i.e., they suffer from a disposition effect, which is a form of regret aversion), they tend to be overconfident (e.g., they are underdiversified and they trade too often), and they exhibit a representativeness bias (i.e., buying recent short-term winners). Both Chinese individual and institutional investors suffer from these behavioral biases, but the former seems more prone to biases than the latter. In comparing our findings for Chinese individual investors with published findings for U.S. individual investors, Chinese individual investors appear just as prone to the disposition effect as U.S. individuals, but Chinese individuals seem to be more overconfident than U.S. individuals. Both of these assessments are consistent with existing research from psychology.

Just as important, we also conduct cross-sectional tests on individual investor behavior. Specifically, we identify middle-aged investors, active investors, wealthier investors, experienced investors, and those from cosmopolitan cities to see whether they are less inclined toward making cognitive errors. Although one might expect these investors to be better investors, as discussed in this study, the existing literature on debiasing has not conclusively shown that experience leads to learned rational behavior. In empirical tests, we find that these investors still exhibit behavioral biases. Overall, our paper shows that emerging market investors make cognitive errors and those individual investors who are commonly considered savvier investors are often not less inclined to make these errors.

Finally, we examine the tendency for individual investors to exhibit multiple psychological biases. Investors who exhibit one type of bias do not necessarily show a propensity for other biases. Indeed, the

¹¹We also conducted a correlation analysis without dummies, in which we use continuous measures of the biases. Overall, the individual investor results are consistent with the results reported in Panel B of Table 8. For institutional investors, some statistically significant correlations was found among measures, but no results were generalizable. These results are available from the authors on request.

presence of some biases is negatively correlated with the presence of others. However, almost 43% of the investors in our sample exhibit more than one bias.

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